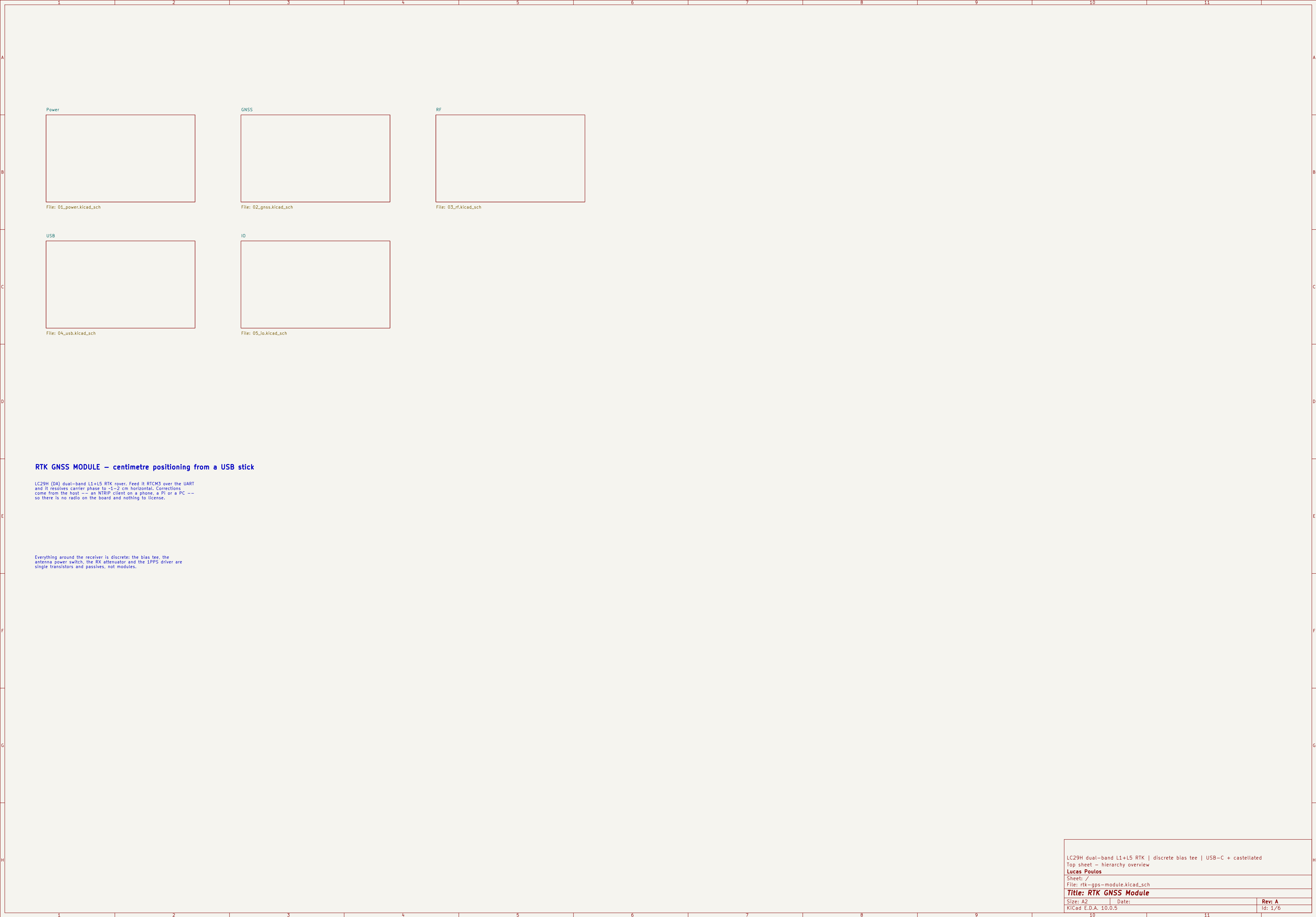




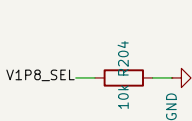
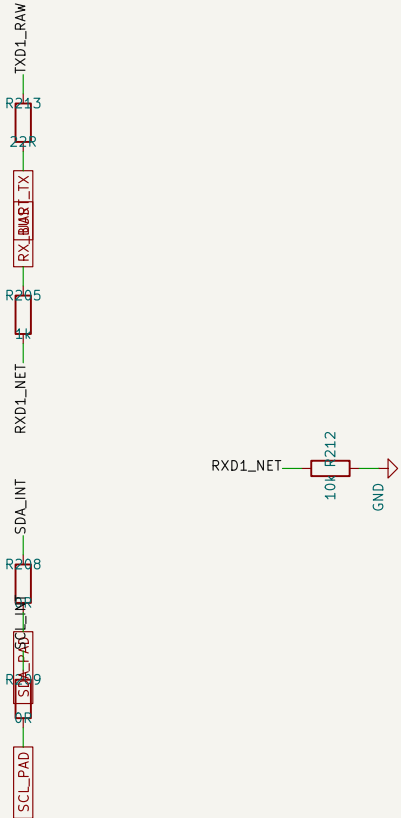
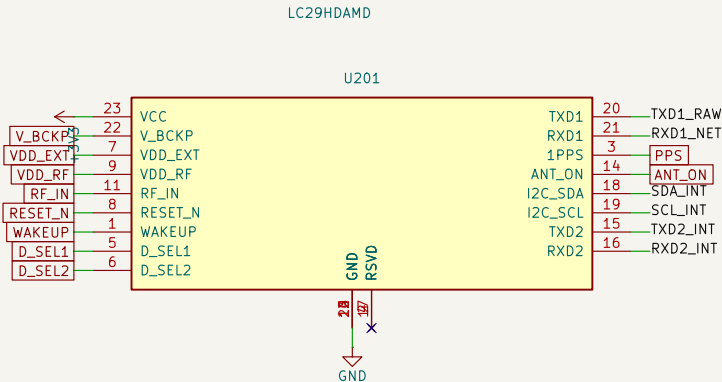
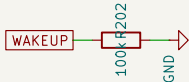
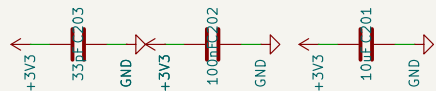
SHEET 1 – INPUT, 5 V ORING, 3V3 AND THE BACKUP DOMAIN

Wired
Sheet 2: Power
Lucas Poulos
Sheet: /Power/
File: 01_power.kicad_sch
Title: USB input, 5 V ORing, 3V3 LDO, backup domain
Size: A3 Date: Rev: A
KiCad E.D.A. 10.0.5 Id: 2/6

Title: USB input, 5 V ORing, 3V3 LDO, backup domain

Size: A3	Date:	Rev: A
KiCad E.D.A. 10.0.5		Id: 2/6

SHEET 2 – LC29H (DA) DUAL-BAND RTK RECEIVER



RX LEVEL. RXD1 is specified VIHmax = 3.08 V (Table 6) and the CH340N drives a full 3.3 V. R205/R212 divide by 10/11 -> 3.0 V, inside spec and still far above the 1.75 V VIHmin. The divider sits AT the module pin, so it protects the castellated RXD1 pad and the USB bridge with one pair of resistors instead of one pair per source. Source impedance 909 R into -15 pF is a 14 ns edge -- negligible against a 1.09 us bit at 921600 baud.

D_SEL1/D_SEL2 pick the host interface. Both are pulled down internally by 75k, and 0.0 selects UART1 -- which is what this board wants, so the default needs no parts. R206/R207 are DNP links to a 1.9 V node for the other three combinations. The 1.9 V comes from dividing VDD_EXT 2.8 V by 4.7k/10k: tying D_SEL to VDD_EXT directly would exceed VIHmax = 2.1 V.

Wired

Sheet 3: GNSS

Lucas Poulos

Sheet: /GNSS/

File: 02_gnss.kicad_sch

Title: LC29H (DA) receiver, decoupling and strapping

Size: A3

Date:

KiCad E.D.A. 10.0.5

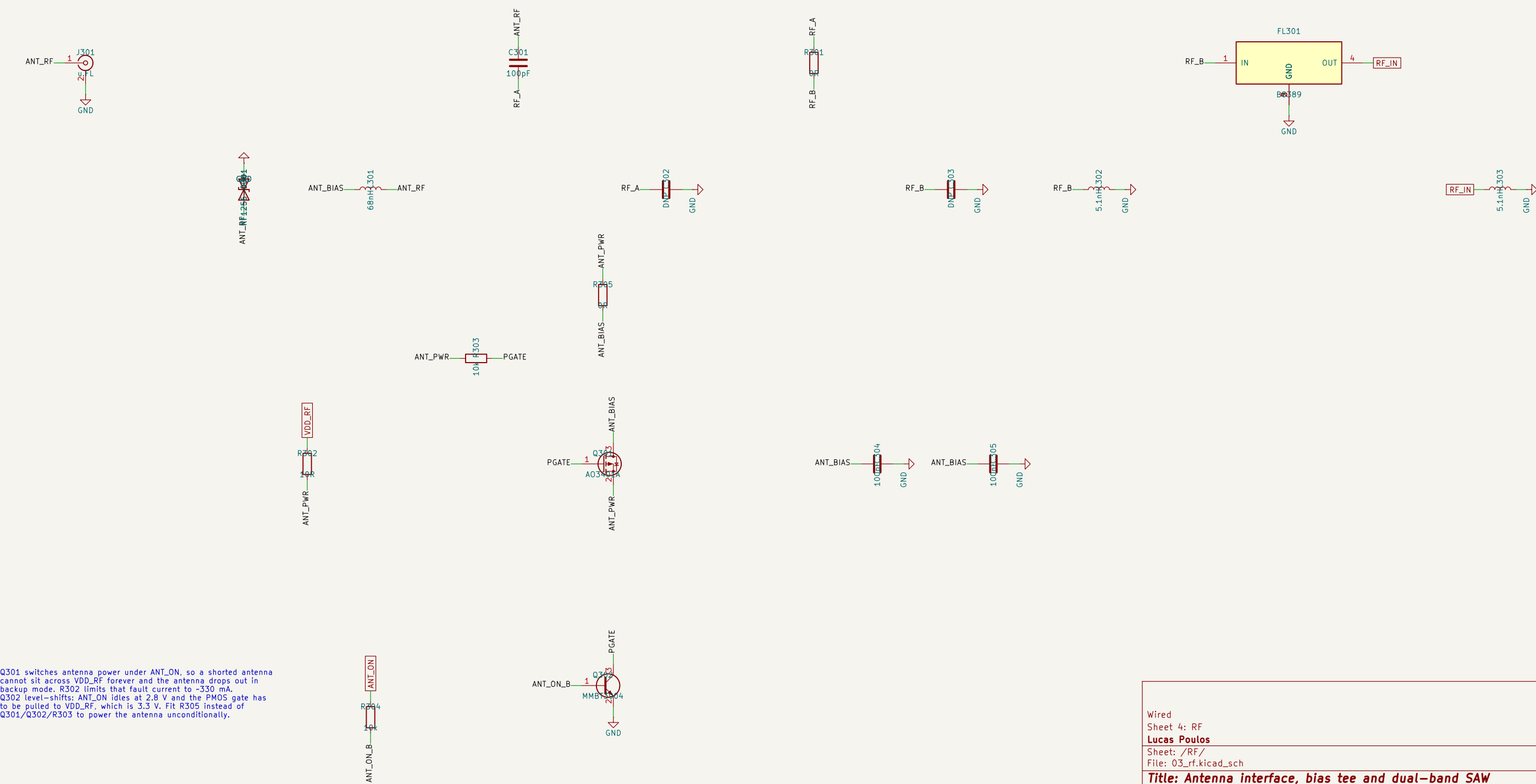
Rev: A

Id: 3/6

SHEET 3 – 50 OHM ANTENNA FRONT END AND BIAS TEE

The whole chain from J301 to RF_IN is a 50 ohm line. L301 injects DC onto it without loading it; 68nH is 673 ohm at L1 and 502 ohm at L5, both high against 50 ohm. C301 blocks that DC out of the module. R301/C302/C303 are a pi network left as 0R + two unfitted pads -- somewhere to retune if a particular antenna misbehaves, at no cost if it does not.

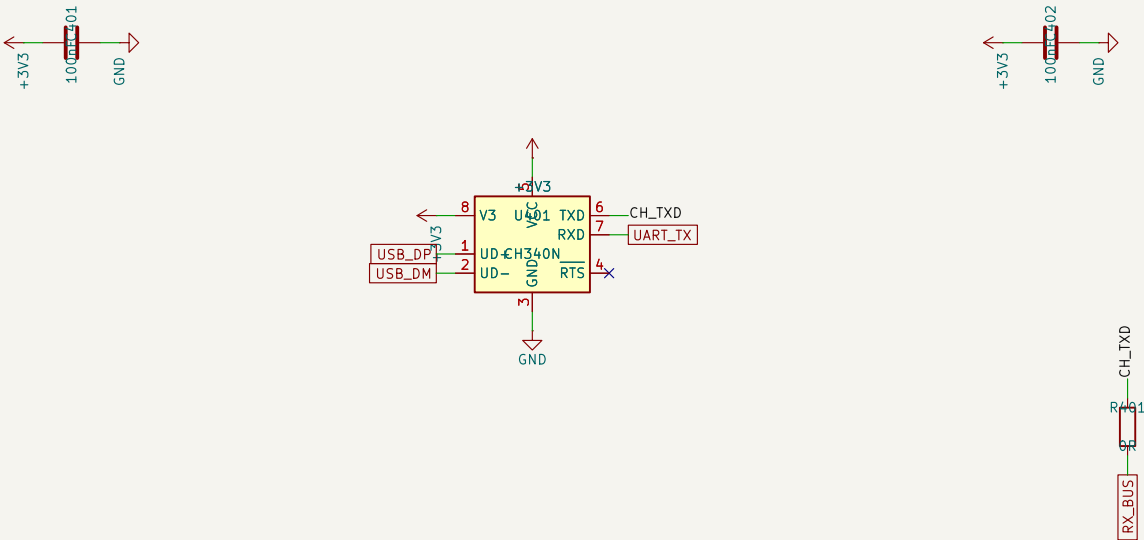
FL301 is the dual-band SAW Quectel names in its own reference design. It passes 1176,45 MHz (L5/E5a/B2a) and 1559-1607 MHz (L1/E1/B1/G1) and rejects 20-39 dB everywhere else. An L1-ONLY SAW here would be a bug: it would delete the second frequency that makes this an RTK receiver rather than a plain GPS. L302/L303 are the 5.1 nH shunts the SAW needs to actually present 50 ohm -- its ports are 50 ohm // 5.1 nH, not 50 ohm.



Q301 switches antenna power under ANT_ON, so a shorted antenna cannot sit across VDD_RF forever and the antenna drops out in backup mode. R302 limits that fault current to ~330 mA. Q302 level-shifts; ANT_ON idles at 2.8 V and the PMOS gate has to be pulled to VDD_RF, which is 3.3 V. Fit R305 instead of Q301/Q302/R303 to power the antenna unconditionally.

Wired		
Sheet 4: RF		
Lucas Poulos		
Sheet: /RF/		
File: 03_rf.kicad_sch		
Title: Antenna interface, bias tee and dual-band SAW		
Size: A3	Date:	Rev: A
KiCad E.D.A. 10.0.5		Id: 4/6

SHEET 4 – USB TO UART BRIDGE

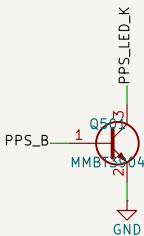
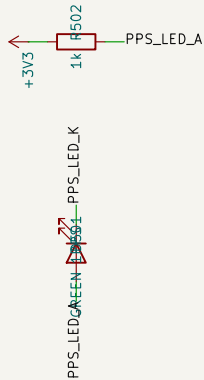
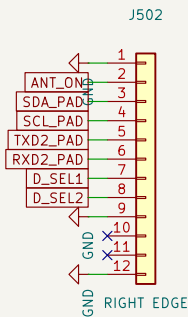
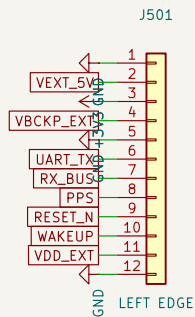


CH340N needs no crystal and costs less than a dollar, which is why it is here rather than a CP2102 or an FT232. Running it at 3.3 V means V3 ties to VCC (WCH datasheet section 5) and the TXD/RXD pair needs no level shifter – only the 3.08 V limit on the receiver’s RXD1, handled by the divider on sheet 2.

R401 is a fitted OR, not a wire. Cut it and the castellated UART pads own the bus outright, so a carrier board’s own host can drive RXD1 without fighting the bridge’s idle-high TXD output. That contention is the one failure mode of having both a USB port and an exposed UART on the same two pins.

Wired		
Sheet 5: USB		
Lucas Poulos		
Sheet: /USB/		
File: 04_usb.kicad_sch		
Title: CH340N USB–UART bridge		
Size: A3	Date:	Rev: A
KiCad E.D.A. 10.0.5		Id: 5/6

SHEET 5 – CASTELLATED EDGE AND INDICATORS



24 castellated pads on 1.27 mm, twelve per edge. Pitch chosen so the module drops onto 0.1 in perfboard and onto the header every carrier already has.
Pads 5–8 on the right edge are the 1.8 V domain (TXD2, RXD2, D_SEL1, D_SEL2). They are NOT 3.3 V tolerant — VIHmax is 2.1 V. Each reaches the edge only through a DNP link so the default board cannot present them to a 3.3 V host by accident.